

platforms like Android, Embedded Linux, OS X, iOS, Windows Phone, etc.

Its features are listed below.

Qt Core:

1. *WinRT*: Support for native settings container.
2. *QStorageInfo*: A class for providing information about currently mounted storage volumes and drives.
3. *QByteArrayList*: Convenience class adding similar extra methods on top of *QList*.
4. *QTabletEvent*: Added information about which stylus buttons are pressed; rotation and proximity events.

Qt Widgets:

1. Improved scaling of widget styles based on DPI.
2. Android style; also works even when not using Ministro for deployment.

Qt QML:

1. *Declarative State Machine (DSM)*: QML wrapper for Qt's State Machine Framework.

Qt Bluetooth:

1. Support for BlueZ5.
2. Preview of Bluetooth Low Energy.

Qt Web Engine:

1. New module with Web engine based on Chromium.

Qt GUI:

1. *QOpenGLContext*: Allows interoperability between Qt and other frameworks.
2. *OpenGL* is improved.
3. *QOpenGLWidget*: Modern replacement for *QGLWidget*.
4. *QRasterWindow*: Class for performing *OpenGL* drawing onto a *QWindow*.
5. *WinRT*: Added support for native file dialogs.

Qt Webkit:

1. Added support for *OpenGL* accelerated 2D canvas to *QtWebKitWidgets*.
2. *OpenGL* acceleration now also works with *QOpenGLWidget*.
3. *Qt Webkit* module is replaced by *Qt WebEngine*.
Qt is available in four editions: Community, Indie Mobile, Professional and Enterprise. The Community edition is available under the open source GNU License 2.1.

Xamarin SDK

Xamarin has created a robust cross-platform mobile application development platform that has been adopted by various IT tech giants like Microsoft, Foursquare, IBM, etc., using Ruby or C#. With the C# shared code base, developers use Xamarin tools to write native Android, iOS and Windows apps with native user interfaces and shared code across multiple platforms.

Xamarin offers two commercial products: Xamarin.iOS and Xamarin.Android. Both are built on top of Mono, an open source version of the .NET framework and can run on various platforms like Linux, UNIX, FreeBSD and MAC OS X.



Figure 5: Xamarin SDK

On iOS, Xamarin's Ahead-of-Time (AOT) compiler compiles Xamarin.iOS applications directly to native ARM assembly code.

On Android, Xamarin's compiler compiles down to the Intermediate Language (IL) which is then Just-in-Time (JIT) compiled to the native assembly when the application launches.

The various products under the Xamarin platform are: Xamarin 2.0, Xamarin.Forms, Xamarin Test Cloud, Xamarin for Visual Studio, Xamarin Studio, Xamarin.Mac, .NET Mobility Scanner, RoboVM, etc.

Xamarin's features are listed below:

1. *Objective-C, Java, C and C++ Interop*: Xamarin provides a platform for directly invoking Objective-C, Java, C and C++ libraries, which provide the unique advantage of existing iOS and Android libraries being written in Objective-C, Java and C/C++.
2. *Modern language constructs*: Xamarin applications are written in C#, a modern language that includes significant improvements over Objective-C and Java such as Lambdas, LINQ, parallel programming features, sophisticated generics, and more.
3. *Binding for SDKs*: Xamarin contains bindings for the entire underlying platform SDKs in both iOS and Android. Being strongly-typed, they are easy to navigate and use, providing robust compile time type checking during any sort of app development.
4. *Cross-platform support*: Xamarin offers cross-platform support for three major application platforms - Android, Windows Phone and iOS. Applications can be written to share up to 90 per cent of their code. 

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